

AI Prompt Logs

Date	15 JUNE 2026
Team ID	XXXXXX
Project Name	Sustainable Waste Management Assistant Using Generative AI
Maximum Marks	3 Marks

AI Prompt Logs:

Maintaining a structured record of your interactions with Large Language Models ensures reproducibility and tracks your prompt engineering iterations. Document your exact prompts, the system context provided, and the resulting outputs to demonstrate how AI accelerated your development phase.

S · N o	Task / Intent	System Prompt & Context	Exact User Prompt	AI Output Summary	Refinements / Iterations
1	Generate API boilerplate	Act as a senior Python developer. Use Flask to build REST APIs.	Generate a Flask boilerplate for waste classification API with routes for /classify and /history.”	AI produced a Flask app with endpoints, basic JSON response, and placeholder functions.	Refined prompt to include Firebase Firestore integration for storing scan history.
2	Prompt Engineering for Waste Classification	Act as an AI prompt engineer. Optimize prompts for	“Design a prompt that classifies waste into categories and provides disposal	AI generated a structured prompt template with placeholders	Iterated to enforce bullet-point outputs and consistent

		Groq LLaMA model.	instructions, hazard warnings, and eco-friendly suggestions.”	for waste item input.	hazard warnings.
3	Dashboard Visualization	Act as a frontend React developer. Use Chart.js.	Generate React code to display a doughnut chart of recyclable vs non-recyclable items.”	AI produced React component with Chart.js integration and sample data.	Refined prompt to include dynamic data from Firestore.
4	Map Integration	Act as a geospatial developer. Use Leaflet.js + OpenStreetMap.	“Generate code to display nearby waste collection centers with color-coded markers	AI produced Leaflet map with markers and popups	Iterated to add filters for hazardous, recyclable, organic, and e-waste categories.
5	Deployment Setup	Act as a DevOps engineer. Use Firebase Hosting.	“Provide steps to deploy React frontend and Flask backend with Firebase Hosting and local server.	AI outlined deployment steps, environment variable setup, and hosting configuration.	Refined prompt to include secure API key management with .env files.